

1 Cycle progression

Overview

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Purpose

The purpose of this overview is to show a timely presentation of a machine's cycle development over a period of time that you can select.

Integration

The data displayed here are collected and saved as part of the machine data collection (MDE). Please also note the information about the database at the end of this section.

Requirements

Please also note the information about the database at the end of this section.

Selection criteria

The application provides the following selection criteria:

Workplace

Enter the number of the workplace/ machine, for which you would like to display a cycle progression.

Point in time ... to ...

When the application is pulled up, the point in time is predefined as follows:

From: Date = "yesterday"/ time = "now"

To: Date = "today"/ time = "now"

Choose the point in time for which the cycle progression should be displayed. Please keep in mind that the length of the point in time will affect data calculation and therefore the response time for the evaluation.

Grid

When the application is pulled up, the grid "point in time" is predefined. Choose the desired grid in which you would like the evaluation to run.



If grid spacing is chosen (not equal to "point in time"), then the calculated actual cycle is the arithmetic mean value of all random samples of actual cycles in the relevant grid spacing period:

$$\overline{IZY} = \frac{1}{n} \sum_{i=1}^n IZY_i$$

The time for the values in this case is the end time of the grid interval. Thus, for example, for hour grids, the values between 13.00 and 14.00 are averaged and 14.00 is displayed as the point in time.

Tabular report

Different presentation options can be chosen for the table view. The below-mentioned data is shown:

Date, time

Point in time when actual cycle data was saved. Please also note the information about the data basis at the end of this section.

Sec/ cycle (depending on the selected table)

Shows the actual cycle in [seconds/cycle].

Cycle/ sec (depending on the selected table)

Shows the actual cycle in [cycles/seconds].

Min/ cycle (depending on the selected table)

Shows the actual cycle in [minutes/cycle].

Cycle/ min (depending on the selected table)

Shows the actual cycle in [cycles/minutes].

LTL (lower tolerance limit)

Calculated lower tolerance limit for the selected machine based on the target cycle available when the actual cycle was saved and on the configuration [Cycle parameter](#).

Formula: LTL = Target cycle - (Target cycle * [Tolerance limit, negative] / 100)

LAL

Calculated lower action limit for the selected machine based on the target cycle available when the actual cycle was saved and on the configuration [Cycle parameter](#).

Formula: LAL = Target cycle - (Target cycle * [action limit, negative] / 100)

UAL

Calculated upper action limit for the selected machine based on the target cycle available when the actual cycle was saved and on the configuration [Cycle parameter](#).

Formula: $UAL = \text{Target cycle} + (\text{Target cycle} * [\text{action limit, positive}] / 100)$

UTL (upper tolerance limit)

Calculated upper tolerance limit for the selected machine based on the target cycle available when the actual cycle was saved and on the configuration [Cycle parameter](#).

Formula: $UTL = \text{Target cycle} + (\text{Target cycle} * [\text{Tolerance limit, positive}] / 100)$



The application does not show the target cycle that is active when saving an actual cycle.

Changes to data in the "machine-related postings" application do not affect this application.

Graphic detail applications

Similar to the tabular detail applications, there are four different detail applications available to show the values as a graphic display, each of which present the data in a different unit:

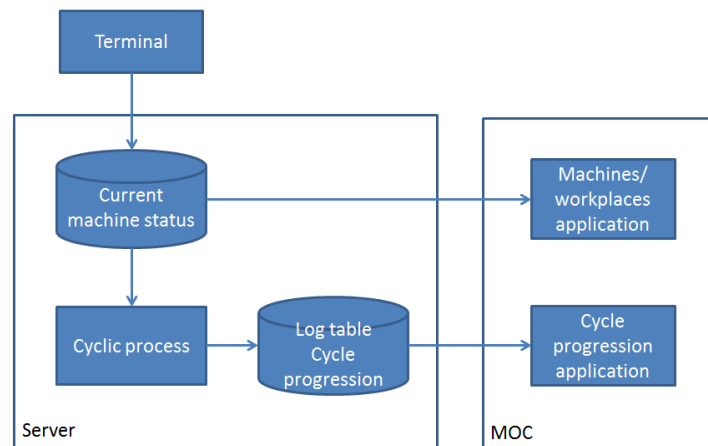
- Seconds/ cycle
- Cycles/ second
- Minutes/ cycle
- Cycles/ minute

The tolerance limits (red) and action limits (yellow) are shown in graphics.

Notes on the data basis for the display of cycle progression

The current actual cycle for each of the separate machines is stored together with the current point in time and the currently set target cycle in a special log table using a cyclic process.

Schematic process:



By default, the cycle for which the process stores the data in the log table for cycle progression is set to every 30 minutes. If necessary, this cycle can also be set to lower intervals (e.g. every 15 minutes) while taking into account the total capacity of the customer's system.



If the value of an actual cycle stays for a longer time at e.g. 0, the scheduler continues to create cyclic entries. All entries include the same actual cycle and the same time stamp (time stamp when the value was set).

Subsequent changes to data in the "machine-related postings" application do not affect the cyclic process.

The application *Cycle progression* accesses values stored in the log table and displays these as a graph in the time progression.

By default, the cycle data for a machine are available for 50 calendar days. If necessary, the data for each machine can also be stored for a longer time (e.g. 90 days) while taking into account the total capacity of the customer's system (must be assured by the customer).

For both cases (modifying logging interval or availability duration of the data), the respective entry must be adjusted in the [Scheduler](#):

Field	Value
Type	S (Standard)
Category	I (interval)
Alterable	Yes
Visible	Visible
Product key	MDE-BP

Field	Value
License key	MDE-BP
HYDRA users	0
Command	./mz_zykl.exe 50
Comment	MDE cycle progression
Interval	00:30:00
Active	☒

You must restart the MES after having modified the values.



The value behind the command is the availability period of log data in days for cycle analyses.